AI-powered Denoising: Supervised Learning

The denoising problem

Estimate this

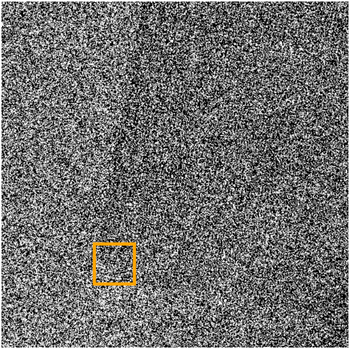
The denoising problem

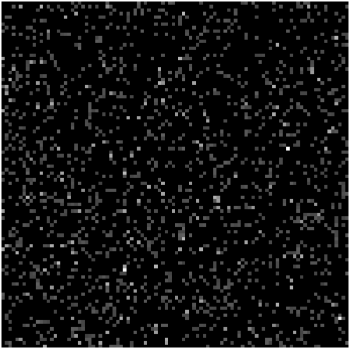
Estimate this

From this

Why?

Why?





How do we evaluate a denoiser?

How do we evaluate a denoiser?

Mean squared error

MSE

$\Big\|$  $-$  $\Big\|^2$

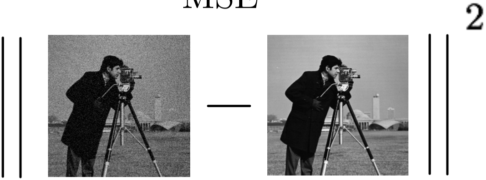
denoised

clean

How do we evaluate a denoiser?

Mean squared error

MSE



denoised clean

Peak signal-to-noise ratio

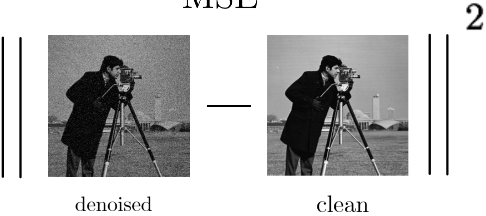
$$\mathrm{PSNR} := 10 \log \left(\frac{\mathrm{M}^2}{\mathrm{MSE}} \right)$$

M: peak signal value

How do we evaluate a denoiser?

Mean squared error

MSE



Peak signal-to-noise ratio

$$\mathrm{PSNR} := 10 \log \left(\frac{\mathrm{M}^2}{\mathrm{MSE}} \right)$$

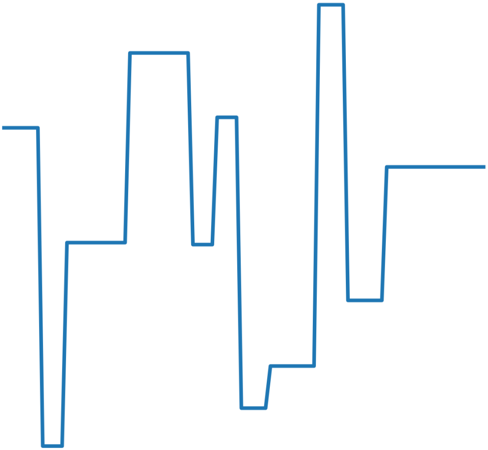
M: peak signal value

Warning: May not reflect application-specific performance

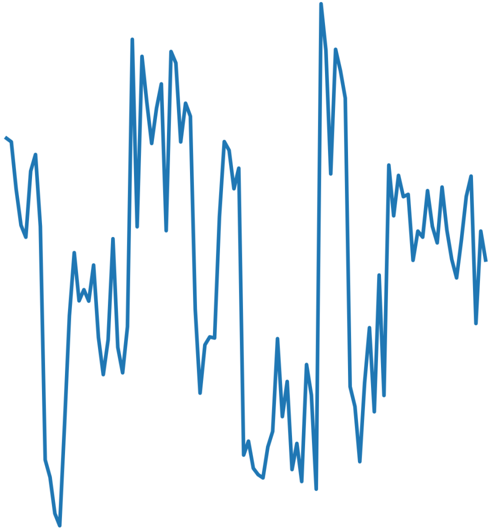
Example datasets

1. 1D piecewise-constant functions
2. Natural photographic images
3. Simulated electron-microscopy data

1D piecewise-constant functions



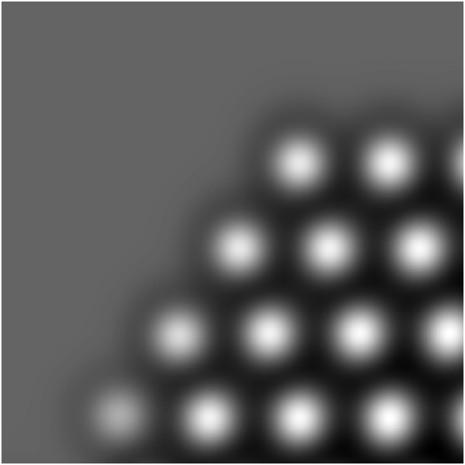
Additive i.i.d. Gaussian noise (PSNR 19.96 dB)



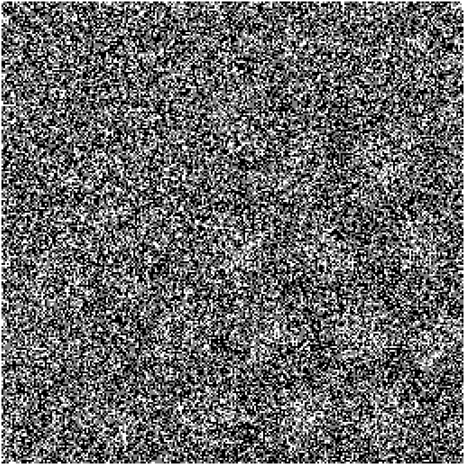
Natural images

Additive i.i.d. Gaussian noise with variable variance (PSNR 19.99 dB)

Electron microscopy



Poisson shot noise (PSNR 6.64 dB)



Plan

Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

Conclusions (and a word of caution!)

Challenge

How to estimate thousands of clean pixels from thousands of noisy pixels?

Challenge

How to estimate thousands of clean pixels from thousands of noisy pixels?

Idea: Assume (1) linearity and (2) shift invariance

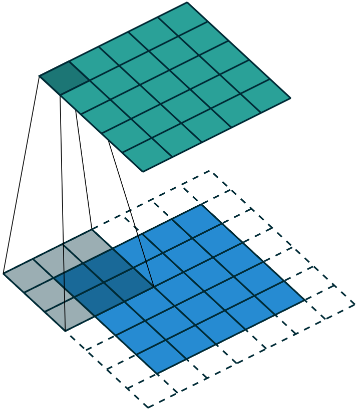
Challenge

How to estimate thousands of clean pixels from thousands of noisy pixels?

Idea: Assume (1) linearity and (2) shift invariance

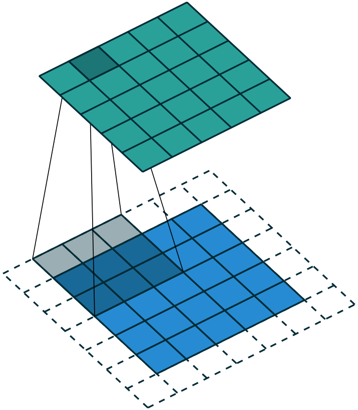
Optimal (Wiener) denoiser is convolutional

Convolutional filter



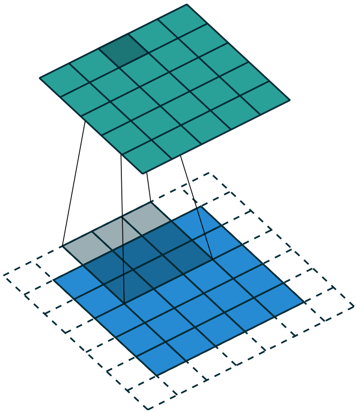
A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Convolutional filter



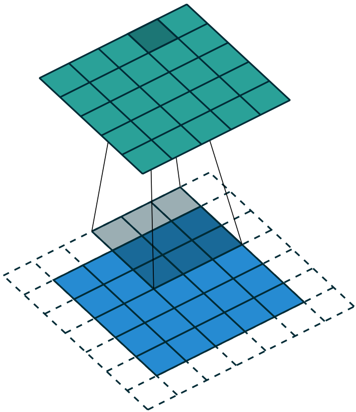
A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Convolutional filter



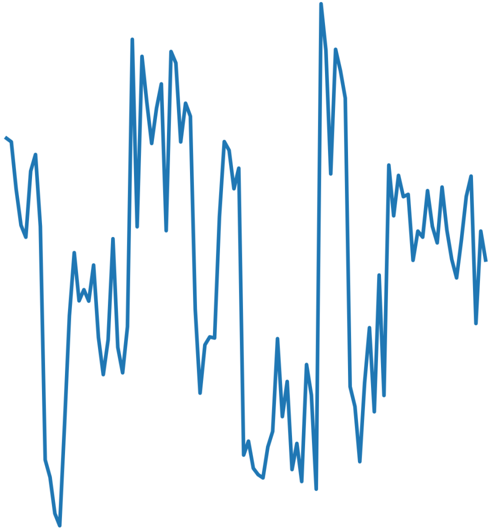
A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Convolutional filter

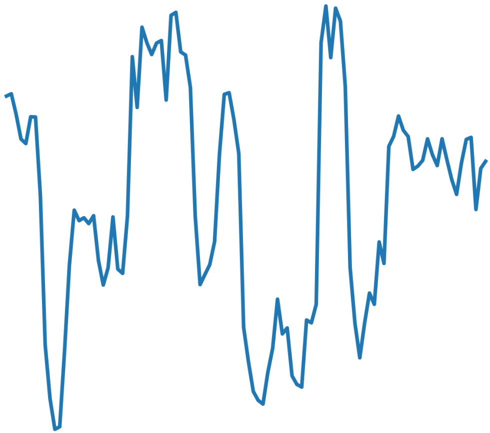


A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Noisy data (PSNR 19.96 dB)



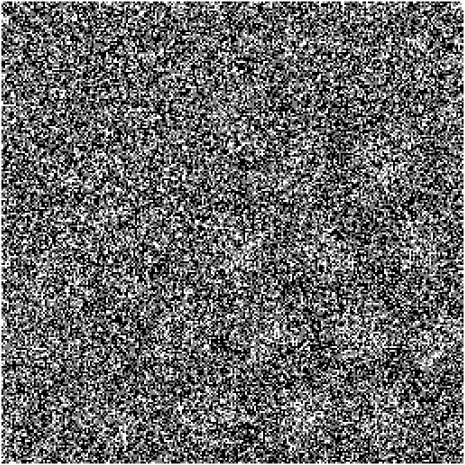
LSI denoising (PSNR 22.33 dB)



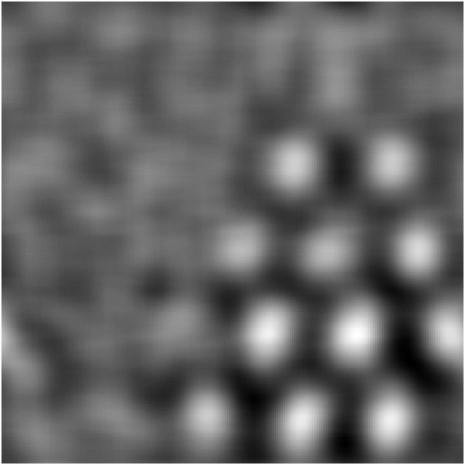
Noisy data (PSNR 19.99 dB)

LSI denoising (PSNR 25.78 dB)

Noisy data (PSNR 6.64 dB)



LSI denoising (PSNR 29.80 dB)



Limitations of LSI denoising?

Limitations of LSI denoising?

Same filter applied at every location

Limitations of LSI denoising?

Same filter applied at every location

Blurs edges and other features

Pre-deep-learning solutions

Pre-deep-learning solutions

- Adapt filter locally (e.g. bilateral filter)

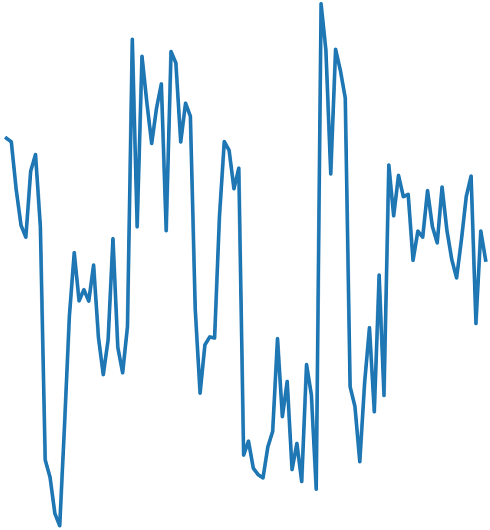
Pre-deep-learning solutions

- Adapt filter locally (e.g. bilateral filter)
- Exploit approximate sparsity of images in transformed domains (finite differences, wavelets...)

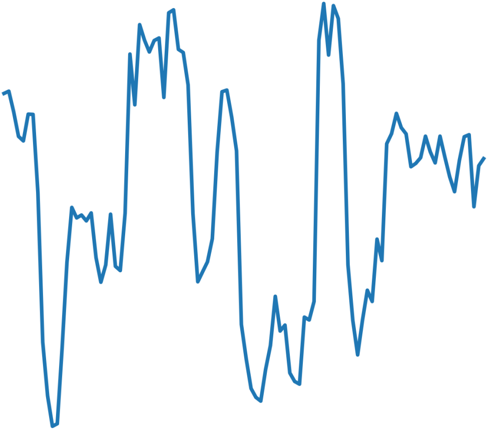
Pre-deep-learning solutions

- Adapt filter locally (e.g. bilateral filter)
- Exploit approximate sparsity of images in transformed domains (finite differences, wavelets...)
- Exploit similarity between image patches to perform nonlocal averaging or collaborative filtering

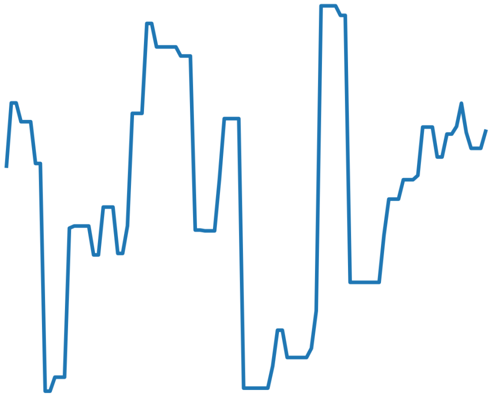
Noisy data (PSNR 19.96 dB)



LSI denoising (PSNR 22.33 dB)



Total variation denoising (PSNR 25.07 dB)



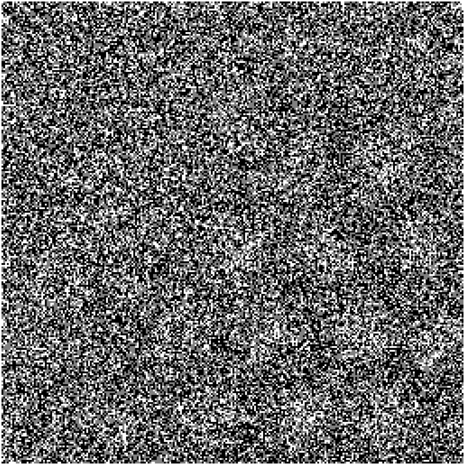
Noisy data (PSNR 19.99)

LSI denoising (PSNR 25.78)

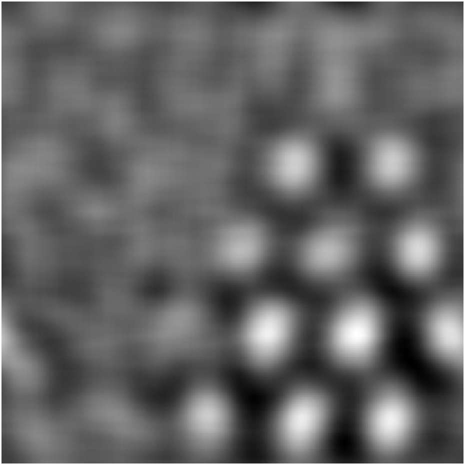
BM3D denoising (PSNR 30.62)

Image denoising by sparse 3-D transform-domain collaborative filtering.
Dabov et al. 2007

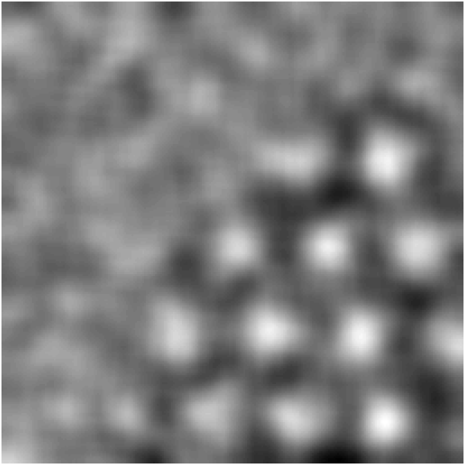
Noisy data (PSNR 6.64)



LSI denoising (PSNR 29.80)



PURE-LET denoising (PSNR 30.72)



A new SURE approach to image denoising: Interscale orthonormal wavelet thresholding. Luisier et al. 2007

Convolutional neural networks

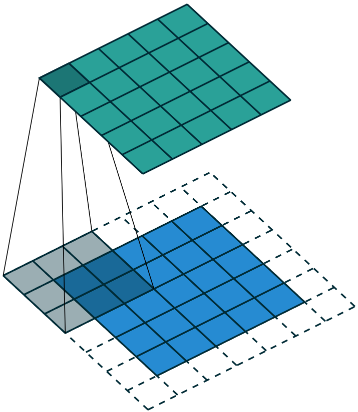
Designed to learn complex nonlinear functions from data

Convolutional neural networks

Designed to learn complex nonlinear functions from data

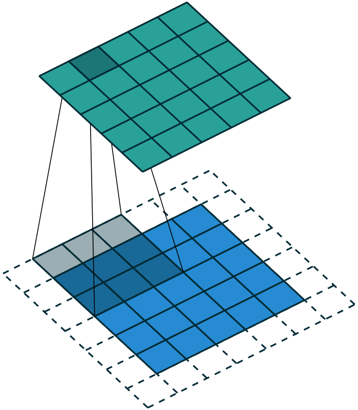
Convolutional layers interleaved with nonlinearities

Convolution



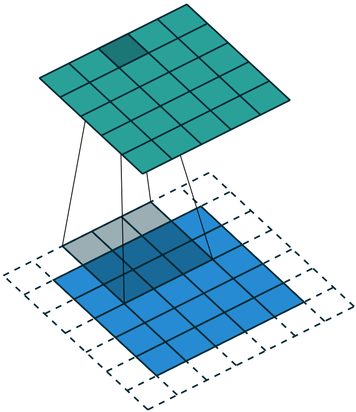
A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Convolution



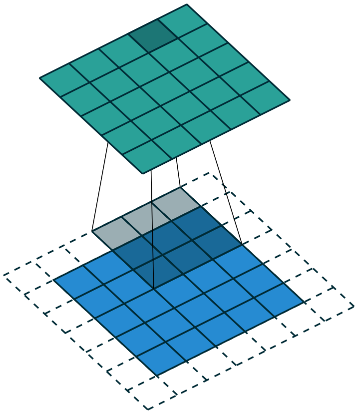
A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Convolution



A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

Convolution



A guide to convolution arithmetic for deep learning, Dumoulin & Visin, 2016

ReLU

Rectifying linear unit

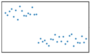
ReLU

Rectifying linear unit

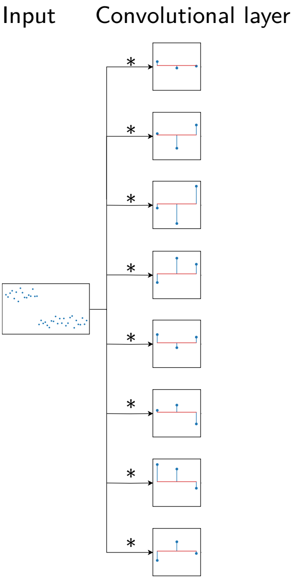
Sets negative entries to zero

Convolutional neural network

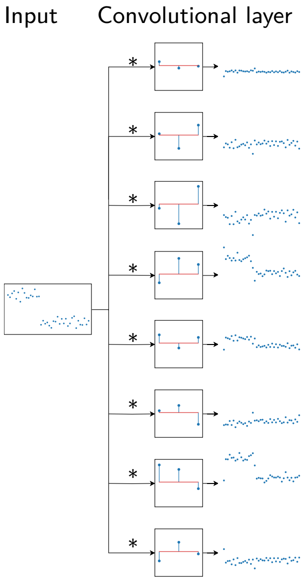
Input



Convolutional neural network

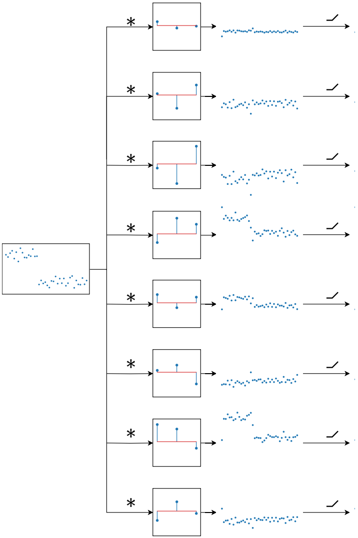


Convolutional neural network

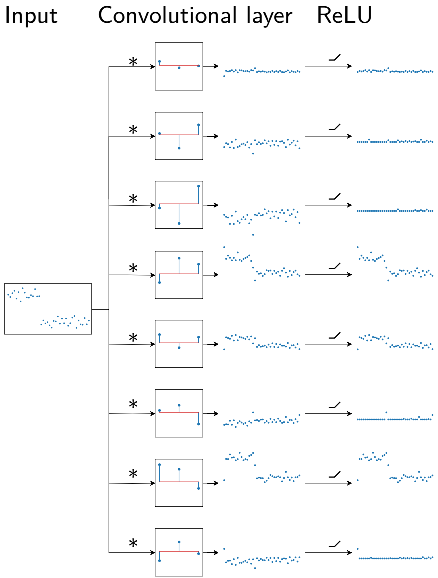


Convolutional neural network

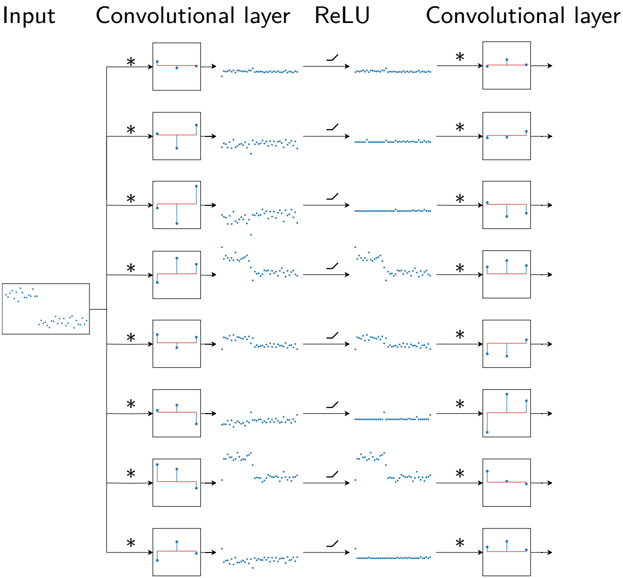
Input Convolutional layer ReLU



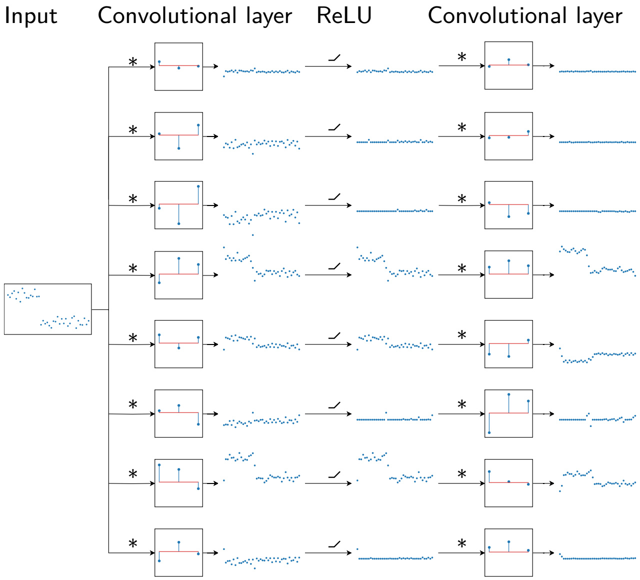
Convolutional neural network



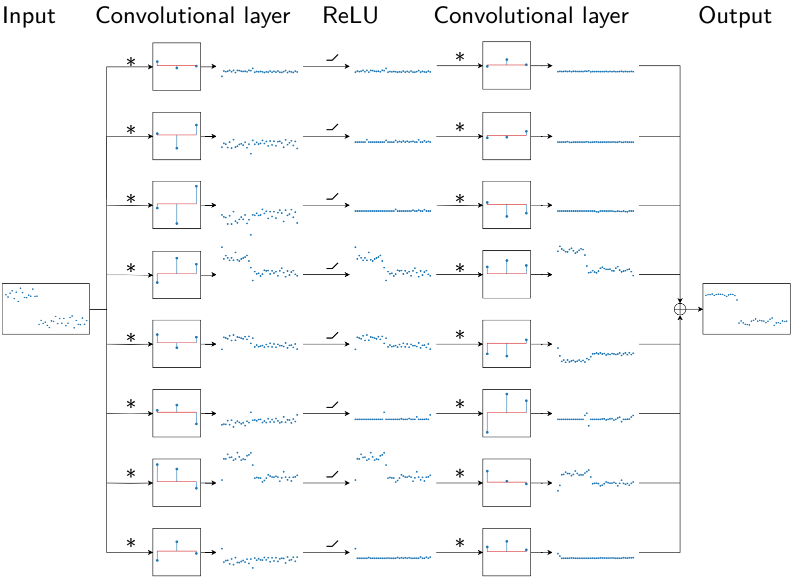
Convolutional neural network



Convolutional neural network



Convolutional neural network



Number of layers?

Number of layers?

Too few layers can result in underfitting

Number of layers?

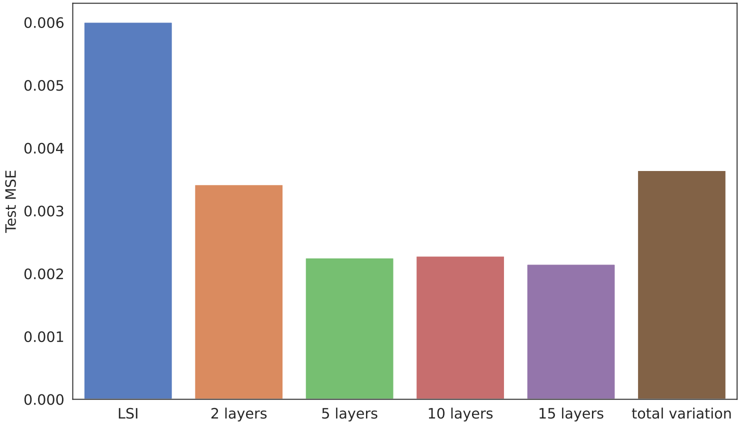
Too few layers can result in underfitting

Once the network is complex enough, diminishing returns

Number of layers?

Too few layers can result in underfitting

Once the network is complex enough, diminishing returns



Piecewise constant signals

Batch normalization

Batch normalization

Features in each channel are

1. Standardized, removing mean and dividing by standard deviation

Batch normalization

Features in each channel are

1. Standardized, removing mean and dividing by standard deviation
2. Re-centered and re-scaled with learned parameters

Skip connections

Skip connections

Links between layers at different depths, bypassing the layers in between

Skip connections

Links between layers at different depths, bypassing the layers in between

In denoising, a skip connection between input and output is often helpful

Skip connections

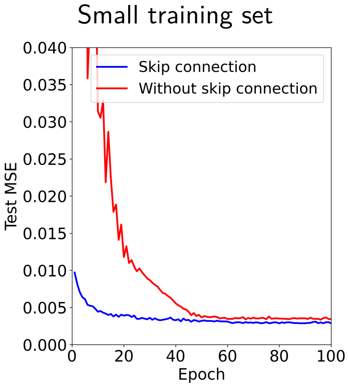
Links between layers at different depths, bypassing the layers in between

In denoising, a skip connection between input and output is often helpful

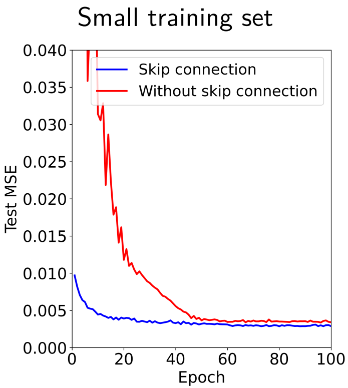
Equivalently, we learn a residual that is added to the noisy input

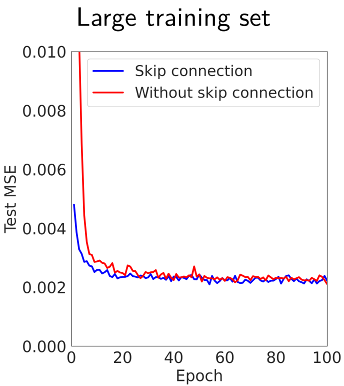
Piecewise constant signals

Piecewise constant signals

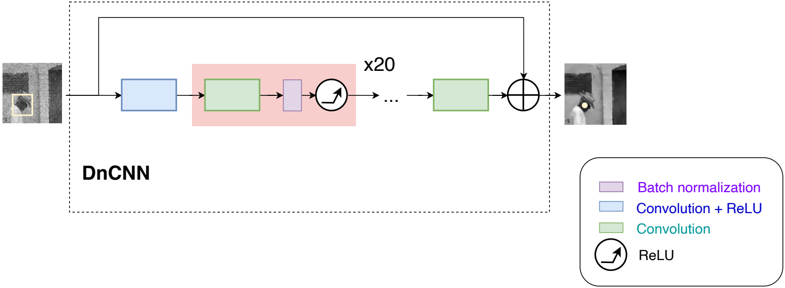


Piecewise constant signals



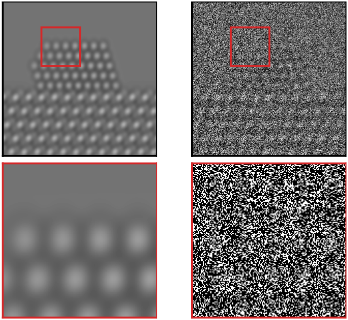


Denoising CNN (DnCNN)



Beyond a Gaussian Denoiser: Residual Learning of Deep CNN for Image Denoising. Zhang et al. 2017

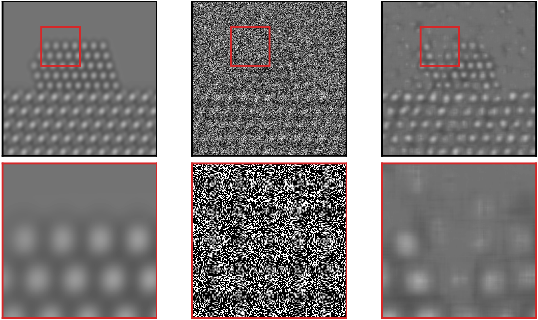
Electron microscopy



Simulated clean image

Noisy image

Electron microscopy

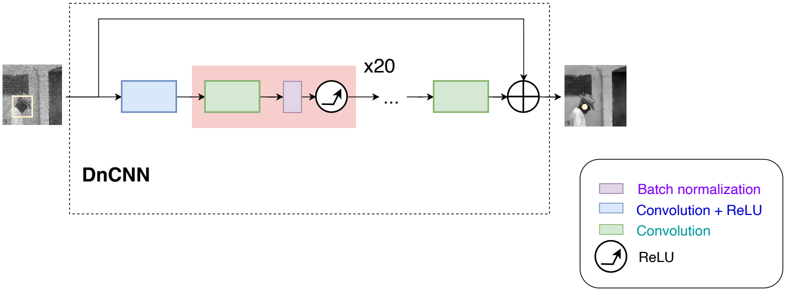


Simulated clean image

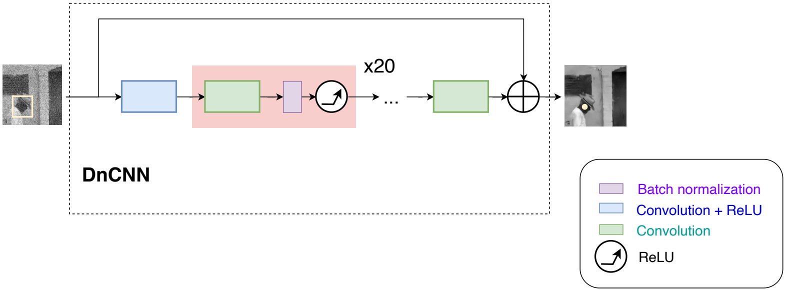
Noisy image

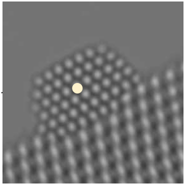
DnCNN

Receptive field

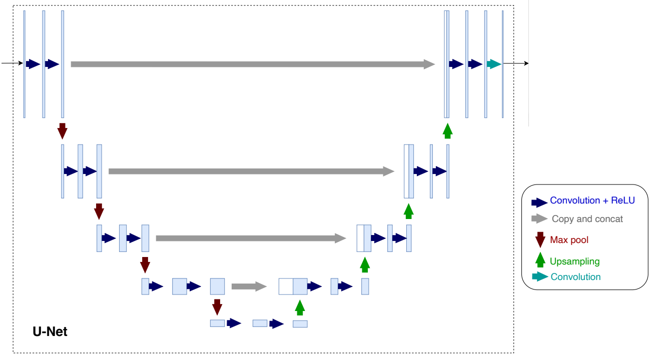


Receptive field



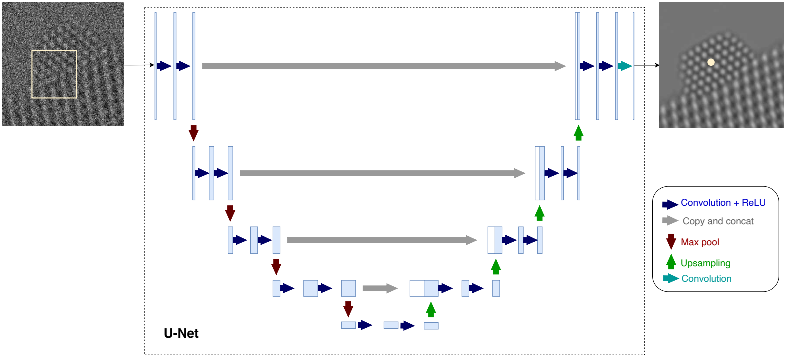


U-Net

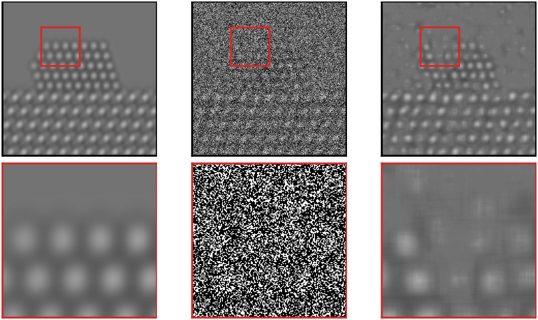


U-Net: convolutional networks for biomedical image segmentation.
Ronneberger et al. 2015

U-Net


U-Net
Convolution + ReLU
Copy and concat
Max pool
Upsampling
Convolution

Electron microscopy

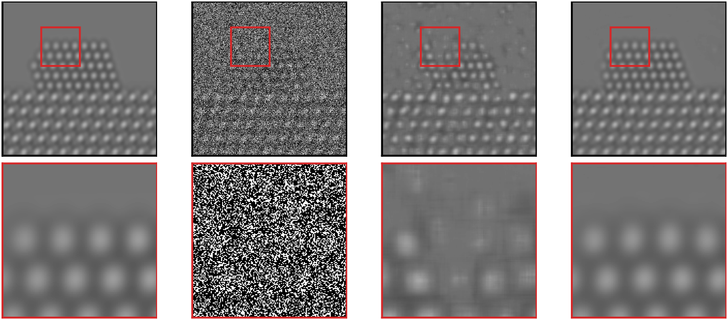


Simulated clean image

Noisy image

DnCNN

Electron microscopy



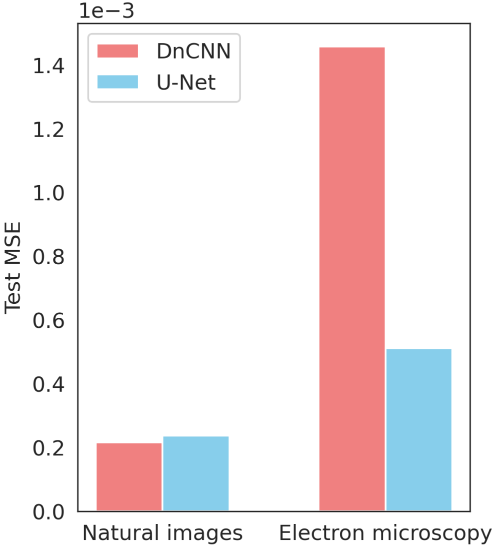
Simulated clean image

Noisy image

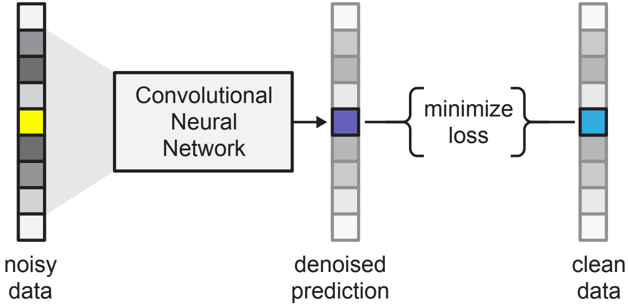
DnCNN

UNet

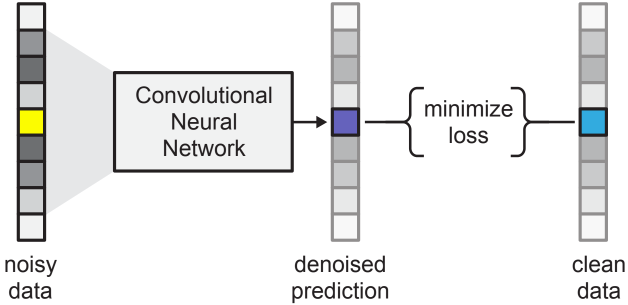
DnCNN vs U-Net



Supervised learning

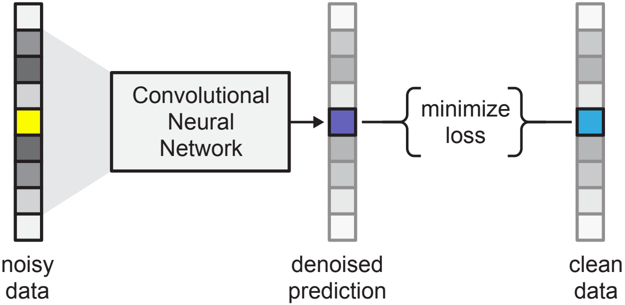


Supervised learning



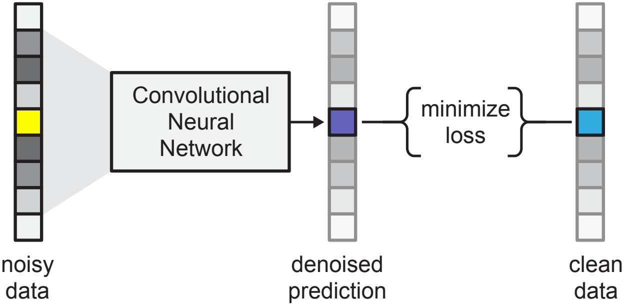
- Assumes access to clean - noisy pairs

Supervised learning



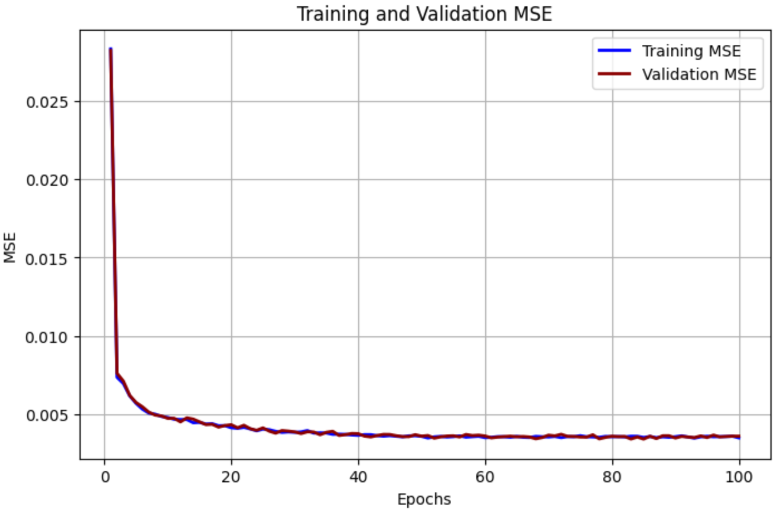
- Assumes access to clean - noisy pairs
- Minimizes mean squared error over training pairs via stochastic gradient descent

Supervised learning



- Assumes access to clean - noisy pairs
- Minimizes mean squared error over training pairs via stochastic gradient descent
- Gradient updates computed via backpropagation

Piecewise constant functions



Continuous noise sampling

If we know the noise distribution, we can simulate it continuously during training

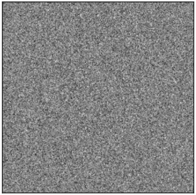
Continuous noise sampling

If we know the noise distribution, we can simulate it continuously during training

Continuous noise sampling

If we know the noise distribution, we can simulate it continuously during training

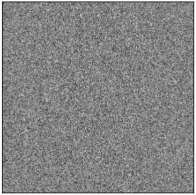
$+$



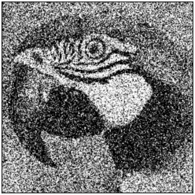
Continuous noise sampling

If we know the noise distribution, we can simulate it continuously during training

$+$

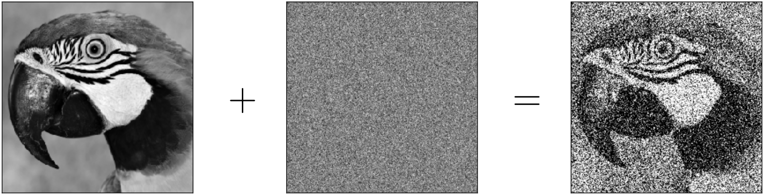


$=$



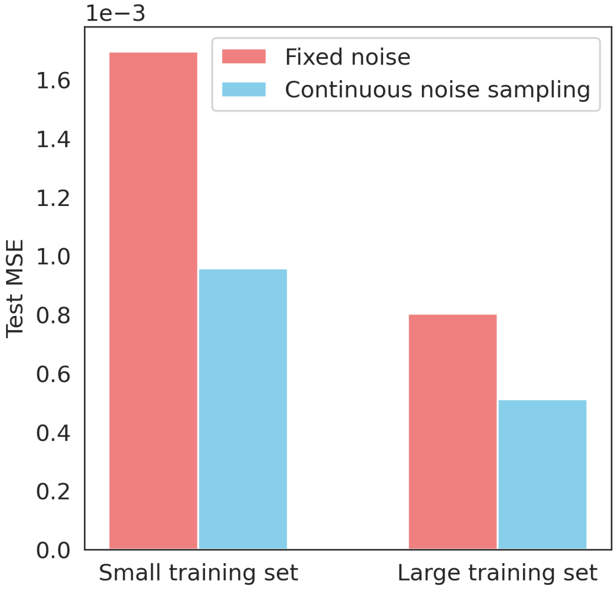
Continuous noise sampling

If we know the noise distribution, we can simulate it continuously during training



Prevents overfitting specific noise samples

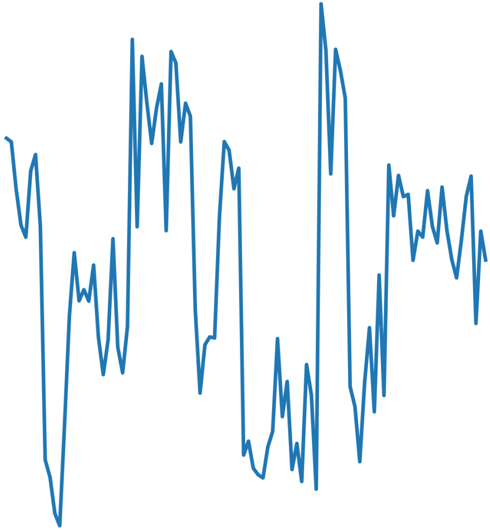
Electron microscopy data



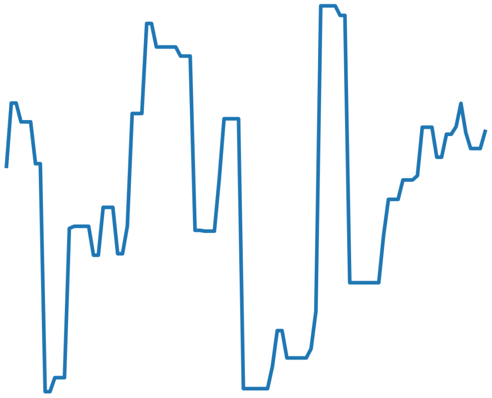
Example datasets

1. 1D piecewise-constant functions
2. Natural photographic images
3. Simulated electron-microscopy data

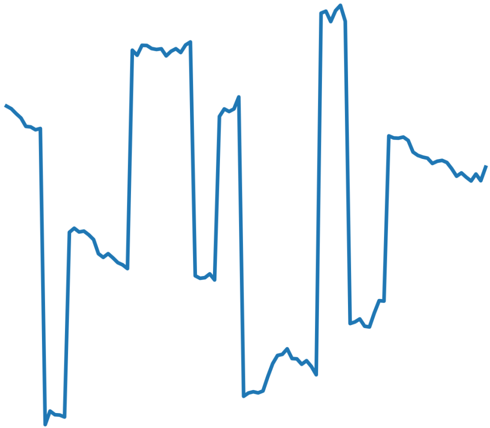
Noisy data (PSNR 19.96 dB)



Total variation denoising (PSNR 25.07 dB)



Neural network (PSNR 26.44 dB)

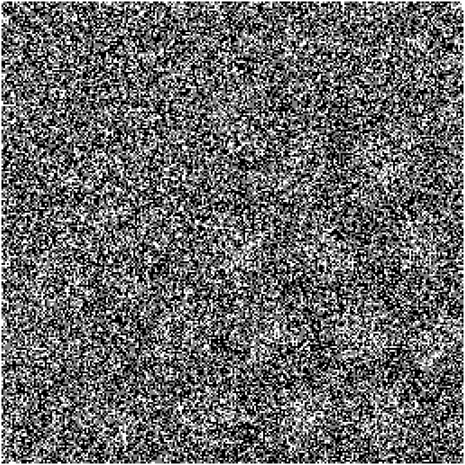


Noisy data (PSNR 19.99 dB)

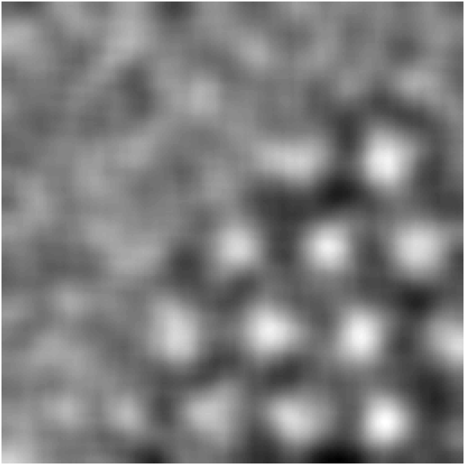
BM3D denoising (PSNR 30.62 dB)

Neural network (PSNR 30.78 dB)

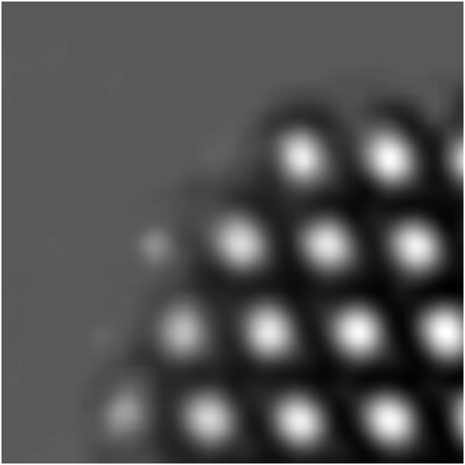
Noisy data (PSNR 6.64 dB)



PURE-LET denoising (PSNR 30.72 dB)



Neural network (PSNR 34.78 dB)



How do neural networks denoise?

They implement complex nonlinear functions that depend on the input

How do neural networks denoise?

They implement complex nonlinear functions that depend on the input

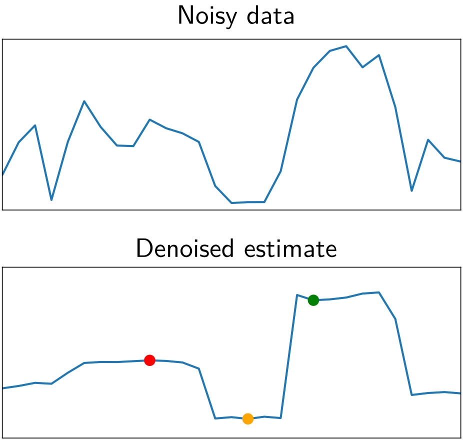
Gradient of output with respect to input reveals learned denoising mechanism *for that specific input*

How do neural networks denoise?

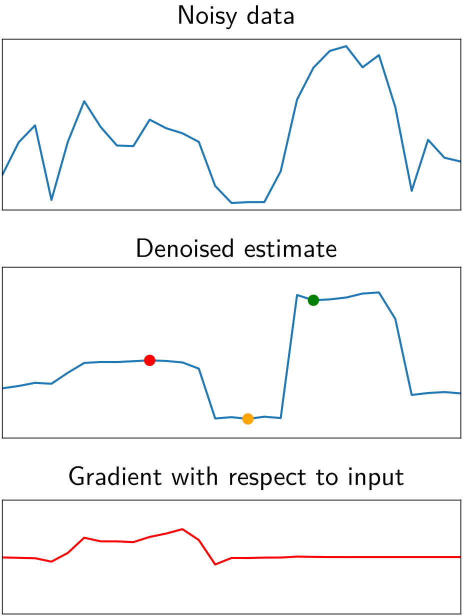
They implement complex nonlinear functions that depend on the input

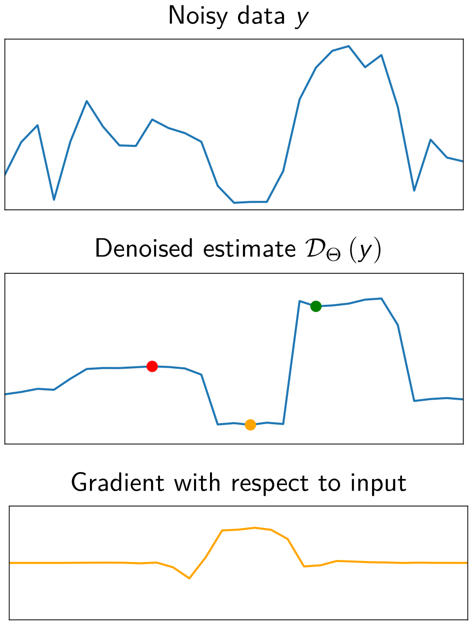
Gradient of output with respect to input reveals learned denoising mechanism *for that specific input*

Robust and interpretable blind image denoising via bias-free convolutional neural networks. Mohan et al. 2020

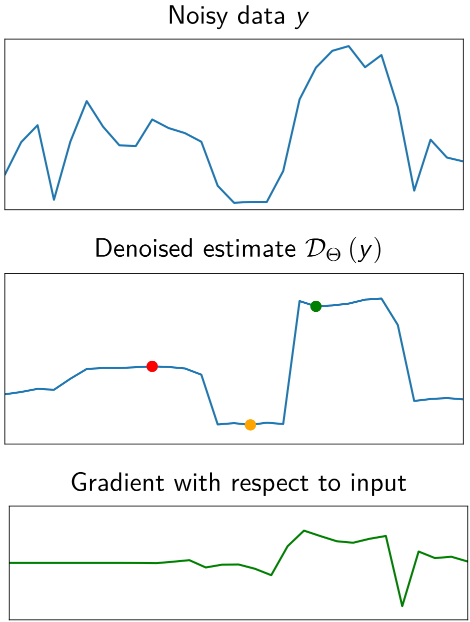


Noisy data
Denoised estimate

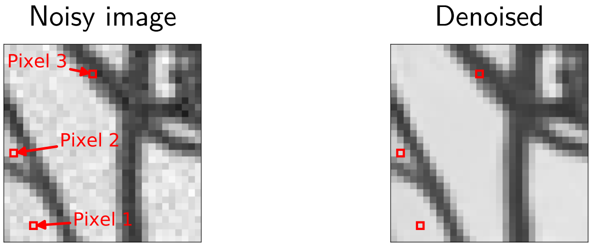

Noisy data
Denoised estimate
Gradient with respect to input



Noisy data y
Denoised estimate $\mathcal{D}_{\Theta}(y)$
Gradient with respect to input

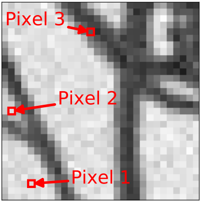

Noisy data y
Denoised estimate $\mathcal{D}_{\Theta}(y)$
Gradient with respect to input

Low noise

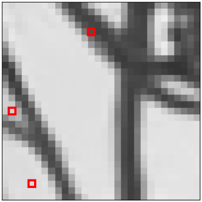


Low noise

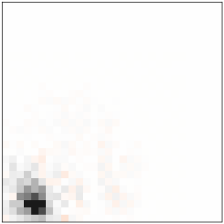
Noisy image



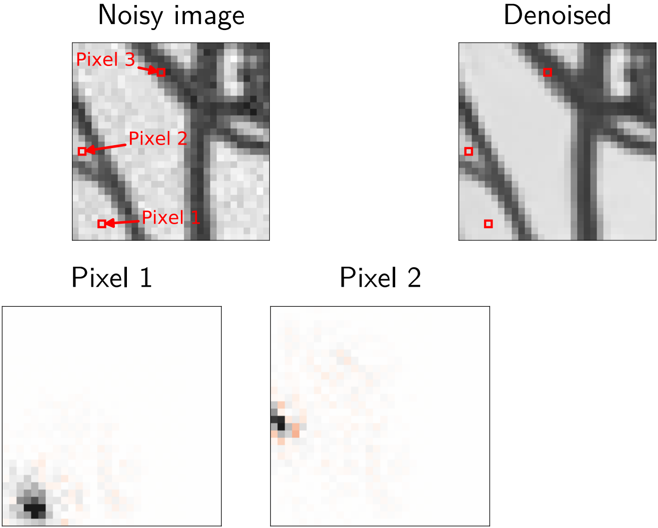
Denoised



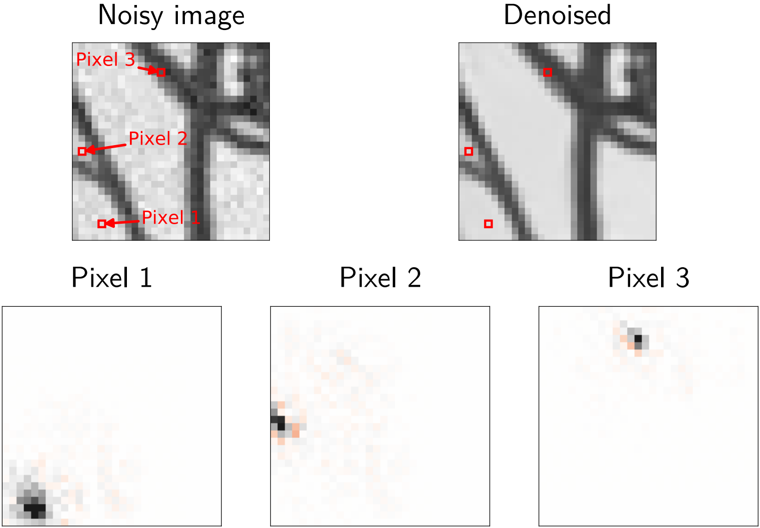
Pixel 1



Low noise

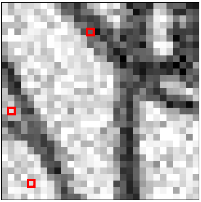


Low noise

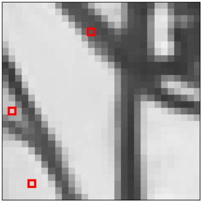

Noisy image
Denoised
Pixel 3
Pixel 2
Pixel 1
Pixel 1
Pixel 2
Pixel 3

Medium noise

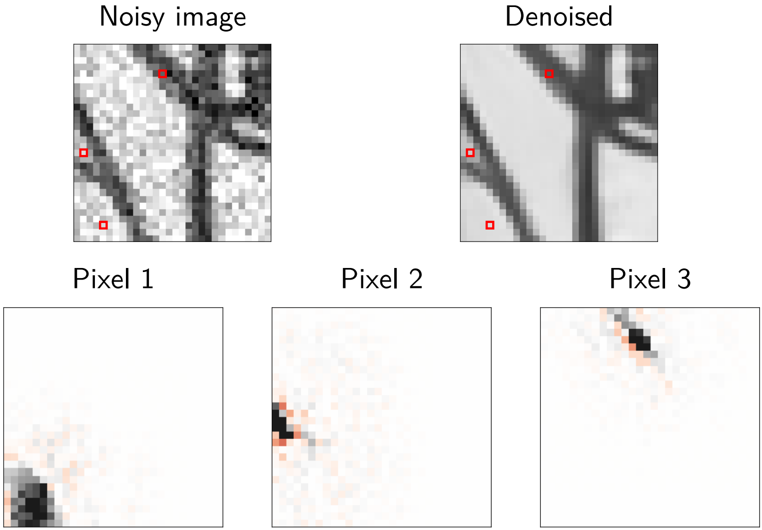
Noisy image



Denoised

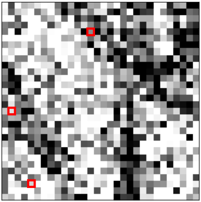


Medium noise

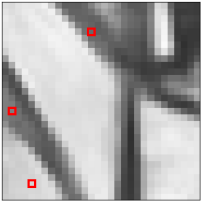


High noise

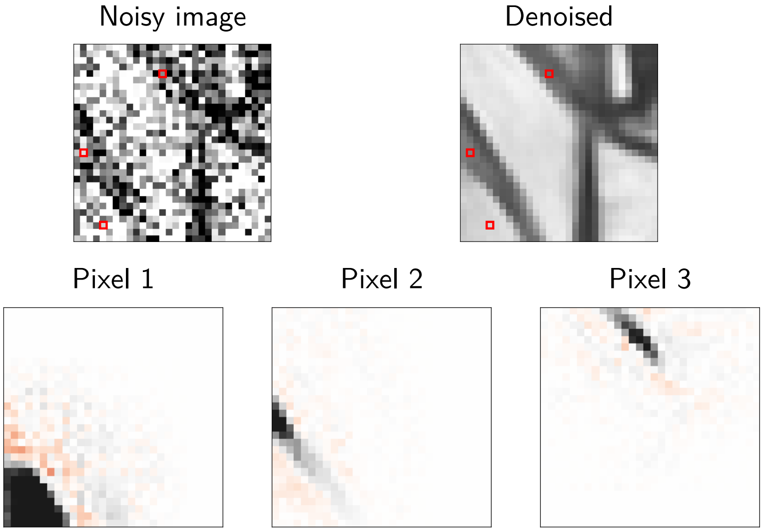
Noisy image



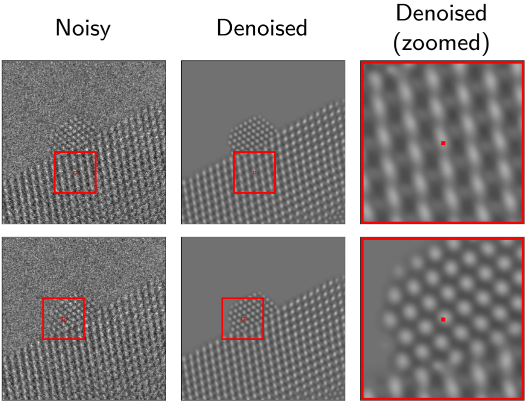
Denoised



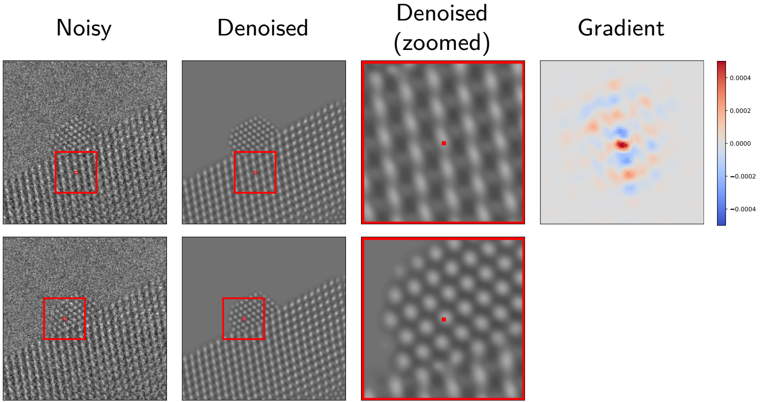
High noise



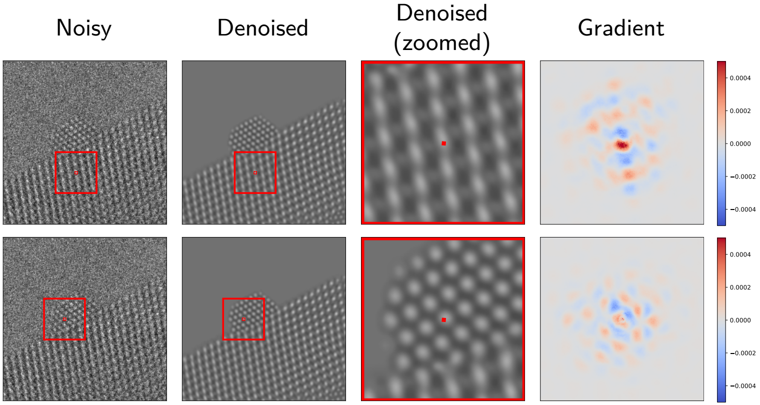
Electron microscopy



Electron microscopy

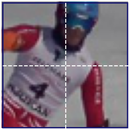


Electron microscopy

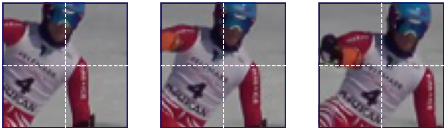


Video denoising

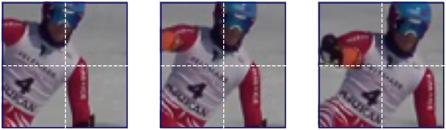
Video denoising



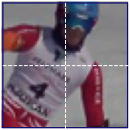
Video denoising



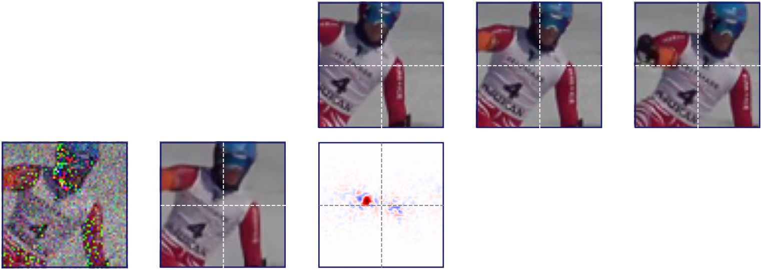
Video denoising



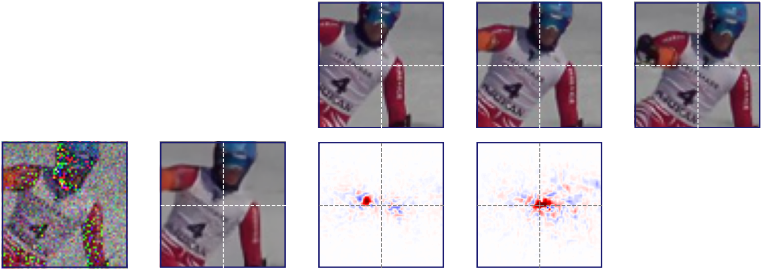
Video denoising



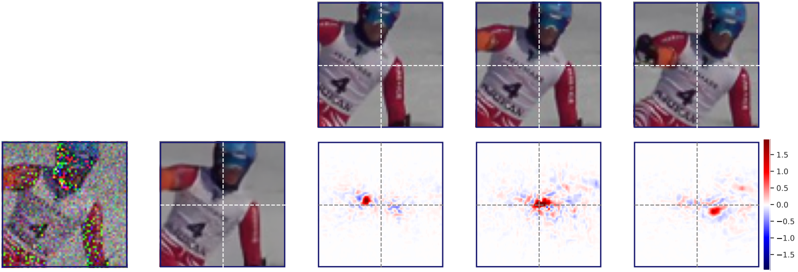
Video denoising



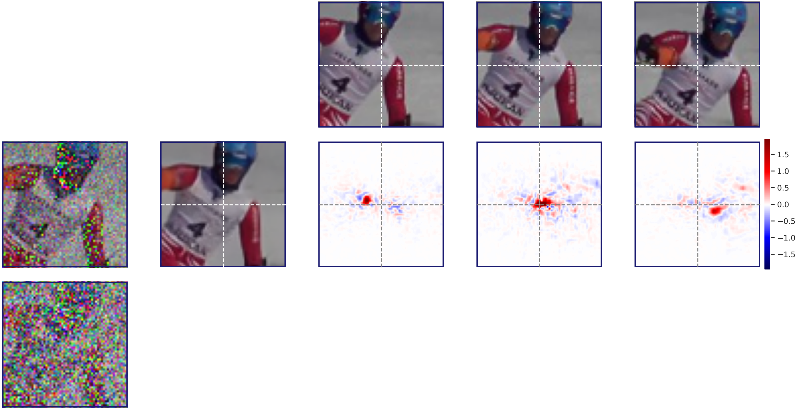
Video denoising



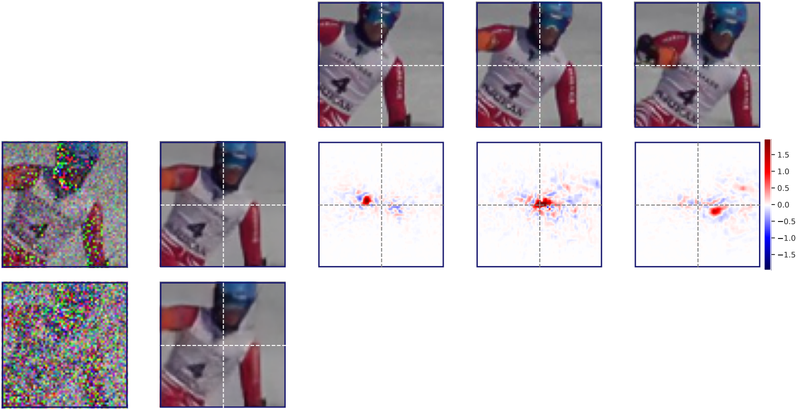
Video denoising



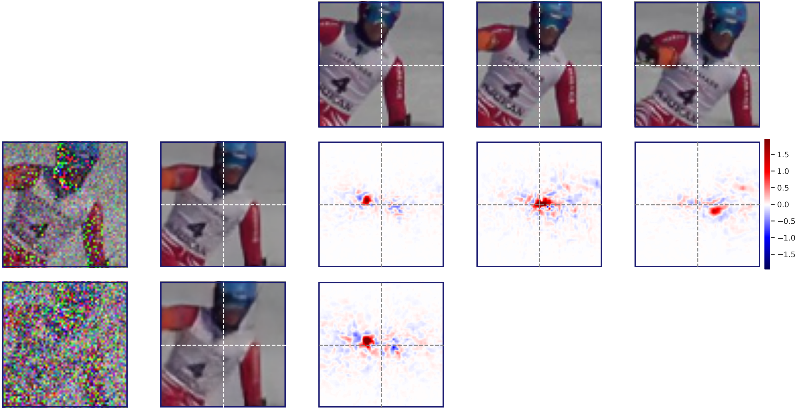
Video denoising



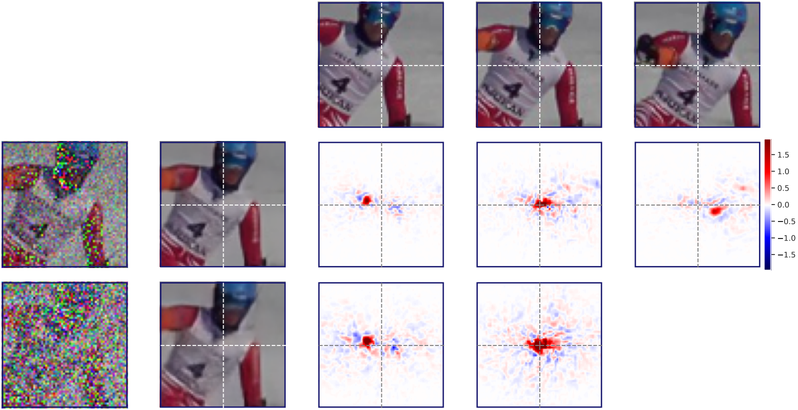
Video denoising



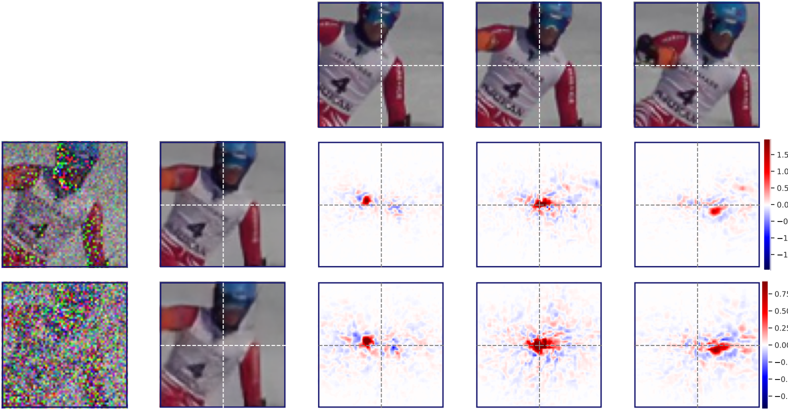
Video denoising



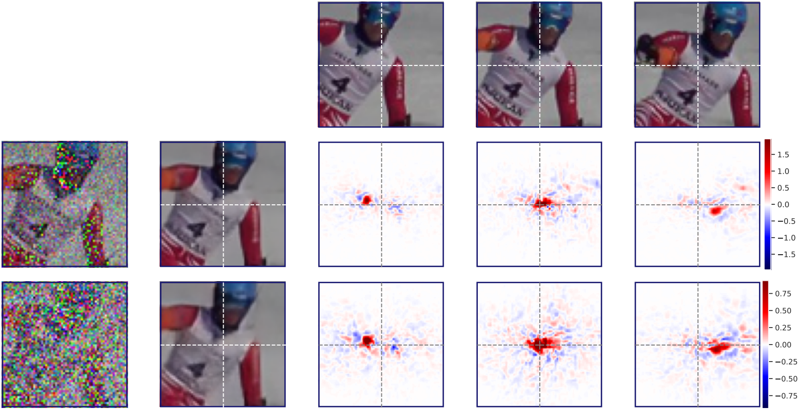
Video denoising



Video denoising



Video denoising



Unsupervised Deep Video Denoising. Sheth et al. 2021

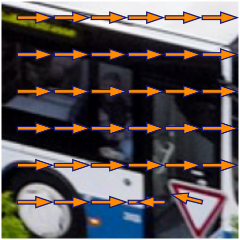
Implicit motion compensation

(a) Noisy frame
$(\sigma = 30)$

Implicit motion compensation

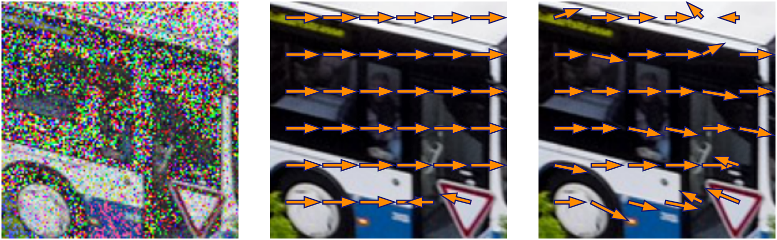
(a) Noisy frame
($\sigma = 30$)

(b) Motion estimate
from clean video



Implicit motion compensation

(a) Noisy frame ($\sigma = 30$) | (b) Motion estimate from clean video | (c) Motion estimate from network gradients



Denoising before deep learning

Convolutional neural networks

Supervised denoising

What do deep denoisers learn?

Conclusions (and a word of caution!)

Conclusions

Conclusions

Supervised denoising is *extremely effective*

Conclusions

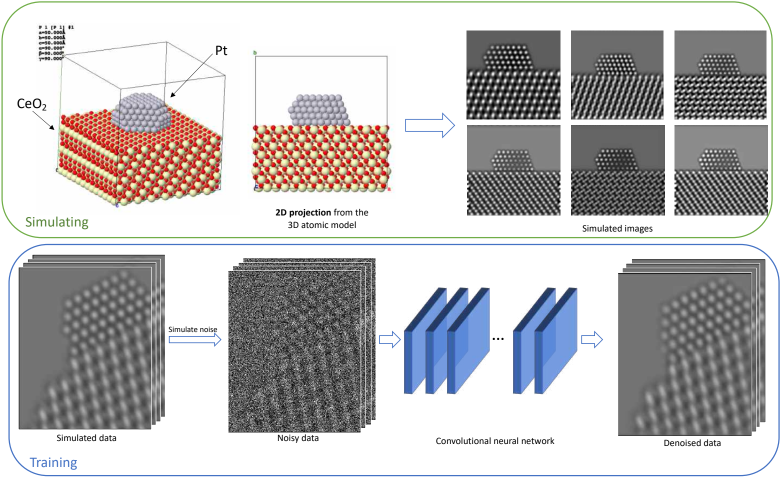
Supervised denoising is *extremely effective*

Models learn denoising strategies adapted to different signals and noise distributions automatically

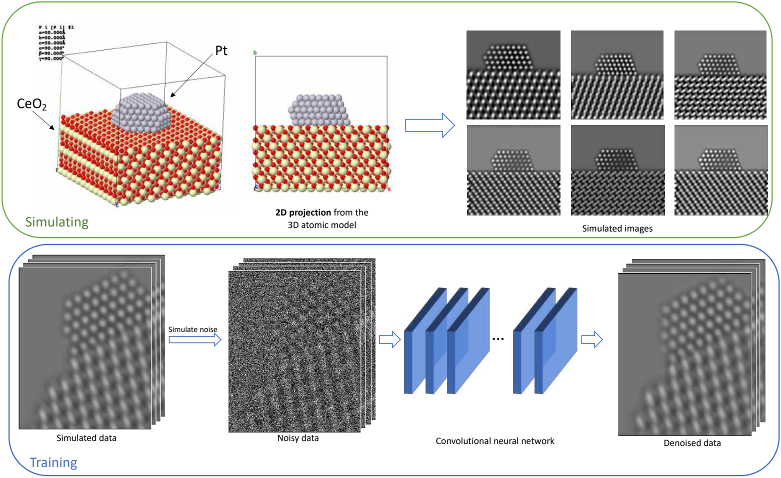
A word of caution

Distribution shift with respect to training data can cause models to fail!

Electron microscopy



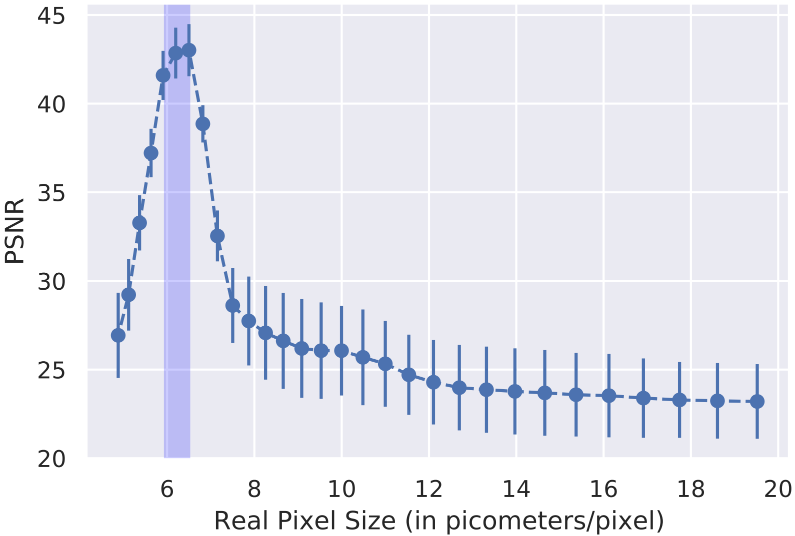
Electron microscopy



Experiment: Apply model to data at a different (1) resolution or (2) orientation

Resolution

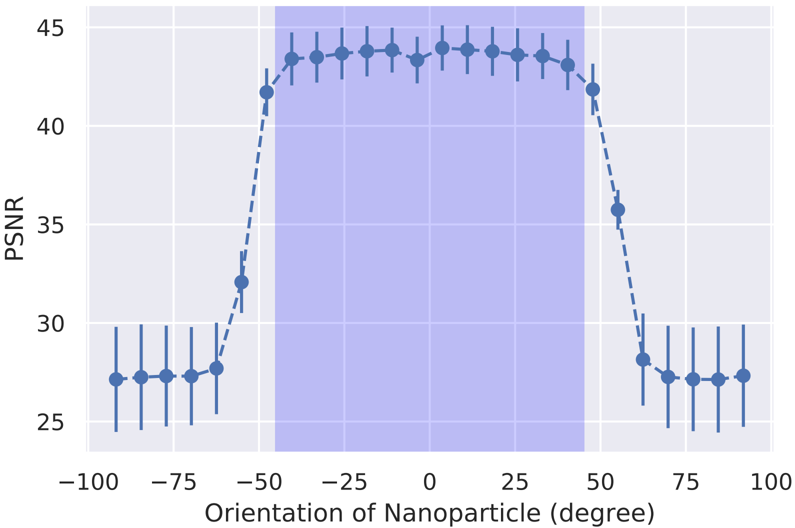
Resolution



45
40
35
30
25
20
PSNR
6
8
10
12
14
16
18
20
Real Pixel Size (in picometers/pixel)

Orientation

Orientation



For more information

Learn, Denoise and Discover: A Guide to Deep Denoising with an Application to Electron Microscopy

Sreyas Mohan, Kangning Liu, Peter Crozier, Carlos Fernandez-Granda

Acknowledgements

We are very grateful for the support of the National Science Foundation through grants OAC-2104105 and OAC-2103936